PROJECT PROPOSAL

# Personal Job Search & Application Agent

One workspace from resume review to interview practice

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# The problem and our solution

### Today

Resumes, job boards, spreadsheets, cover letters and interview study live in separate tools. Context is lost, applications are generic and follow-ups are missed.

### Career Desk

A web app where each stage feeds the next: structured profile, job fit, tailored cover letter, application tracking and adaptive interview prep.

# What the app does

### Resume to profile

Parses PDF or DOCX and extracts experience, education and skills.

### Find and fit

Search roles, then get matched skills, gaps and a fit summary.

### Apply

Tailored cover letters and a tracker from saved to offer.

### Interview prep

Multi-day plans, generated questions, scored answers.

### Career resources

RAG answers grounded in ingested study material.

### Extensions

Semantic job search and stale-application alerts.

# Architecture

### Browser UI

Static HTML, CSS and JavaScript served by FastAPI. No build step.

### FastAPI API

Routers handle HTTP; services hold business logic.

### Language model

One shared client for extraction, writing, questions and scoring.

### PostgreSQL

SQLAlchemy models and Alembic migrations.

### Chroma

Embedded resource chunks and semantic job postings.

### Integrations

Adzuna, Tavily, and a demo job catalog for development.

# Data model

| Table | Holds |
| --- | --- |
| users | Sign-in identity |
| profiles, work_experiences, education, skills | Structured candidate profile |
| applications | Status, cover letter, job link |
| skill_gap_reports | Matched and missing skills per role |
| practice_sessions, practice_answers | Interview plan, scores, feedback |

Resume files are stored outside the database and referenced by path.

# Work plan: eight phases

**1 Foundations**
Models, auth, tracking

**2 Resume**
Parsing and AI extraction

**3 Job search**
Filters, board client

**4 Fit**
Skill gaps, cover letters

**5 Resources**
Embeddings, RAG

**6 Interview**
Plans, scoring

**7 Extensions**
Semantic search, follow-ups

**8 Interface**
Career Desk UI

# Scope, risks and success

### Limits

Demo catalog results are examples, not live vacancies. No profile editing or resume export yet.

### Risks

Extraction errors: user reviews the profile. API outages: mock catalog fallback. Secrets stay in .env.

### Success

A new user completes the full journey without the API docs page, with private per-user data.